Problems [829]

Ch # 28.

Problems 1, 6, 7, 10, 14, 17, 4.

Megnetic force on the charged particle, F_B

$F_B = q\vec{V} \times \vec{B} \Rightarrow qVB\sin\theta$

θ is the angle b/w V and B.

1.696×10^{-12}

1690

10.1556×10^{14}

Example

Given Data

$B = 1.2\,mT$

$K.E = 5.3\,MeV$

$mass = 1.67 \times 10^{-27}\,kg$

$a = ?$

Sol

$K.E = \frac{1}{2} m v^2$

$v^2 = \frac{K\,2}{m}$

$v = \sqrt{\frac{2K}{m}} \Rightarrow \sqrt{\frac{2 \times 5.3 \times 1.60 \times 10^{-13}}{1.67 \times 10^{-27}}}$

$= \sqrt{\frac{1.696 \times 10^{-12}}{1.67 \times 10^{-27}}} \Rightarrow \sqrt{1.0155 \times 10^{15}}$

$= 3.2 \times 10^{7}$

$F_B = qvB\sin\theta$

$= 1.6\times10^{-19} \times 3.2\times10^{7} \times 1.2\times10^{-3}$

$= 6.144\times10^{-15}$

$= 6.1\times10^{-15}$

$F_B = ma$

$a = \frac{F_B}{m} \Rightarrow \frac{6.1\times10^{-15}}{1.67\times10^{-27}}$

$\boxed{a = 3.7\times10^{12}\ m/s^{-2}}$

$\boxed{V = \frac{E}{B}}$

Crossed Fields, Hall Effect

$eE = eV_dB$

$J = neV_d$

$V_d = \frac{J}{ne} \Rightarrow \frac{i}{neA}$ $\quad \because \boxed{J = \frac{i}{A}}$

$V_d = \frac{J}{ne}$

$eE = e\frac{\sigma i B}{nqA}$ $\quad n = \frac{\sigma}{}$

$\boxed{n = \frac{Bi}{Vle}}$ $\quad \boxed{l = \frac{A}{d}}$

Circulating Charged Particle

$$F = \frac{mv^2}{r} \quad , \quad F = qvB\sin\theta$$
$$= qvB$$

1) $$qvB = \frac{mv^2}{r}$$

$$r = \frac{mv^{\cancel{2}\,1}}{q\cancel{v}B} \Rightarrow \frac{mv}{qB}$$

2) $$T = \frac{2\pi r}{v} \Rightarrow \frac{2\pi}{\cancel{v}} \frac{m\cancel{v}}{qB} \Rightarrow \frac{2\pi m}{qB}$$

3) $$f = \frac{1}{T} = \frac{qB}{2\pi m}$$

4) $$\omega = 2\pi f \Rightarrow \cancel{2\pi} \frac{qB}{\cancel{2\pi} m}$$

$$\boxed{\omega = \frac{qB}{m}}$$

Example 28·02

Given Data

$d = 1.5\,cm$

$v = 4\,ms^{-1}$

$B = 0.050\,T$

$V = ?$

Sol

$qE = qvB$

$E = vB \rightarrow (i)$

$V = Ed$ — (ii)

$\therefore V = vBd$

$\Rightarrow 4 \times 0.05 \times 0.015$

$V = 3 \times 10^{-3}\,V.$

$\boxed{V = 3\,mV}$

Helical Paths

Velocity vectore resolved into 2 components

1) $v_{\parallel} = v\cos\phi$ Parallel

2) $v_{\perp} = v\sin\phi$ Perpendicular

7.903×10^{12}
$= 2.81\times10^{6}$

$\frac{45}{\sqrt{4.9\times10^{-31}}}$

Example 3

Given Data

mass of electron $= 9.11\times10^{-31}$

$e = 1.67\times10^{-19}$

$K.E = 22.5\,eV$

$B = 4.55\times10^{-4}\,T$

$\theta = 65.5^\circ$

Pitch $= P = ?$

Sol:

$K = \frac{1}{2} mv^2$

$v = \sqrt{\frac{2Ke}{m}} \Rightarrow \sqrt{\frac{2\times22.5\times1.6\times10^{-19}}{9.11\times10^{-31}}}$

$\boxed{v = 2.81\times10^{6}}$

0.4146

$P = v_{\parallel} \cos\theta\, T \times \cancel{?}$

$= v_{\parallel} \cos 65.5 \times \frac{2\pi m}{qB}$

$= 2.81\times10^{6} \cos 65.5 \times \frac{2\times3.14\times9.11\times10^{-31}}{1.6\times10^{-19}\times4.55\times10^{-4}}$

$= 160.76\times0.4146\times10^{-31+19+4}$

7.28

$= 9.1553\times10^{-8}$

$P \boxed{= 9.16\ cm}$

Example 6

Given Data

$i = 28\ A$

$m = 46.6\ g/m \Rightarrow \frac{46.6}{1000} \Rightarrow 0.0466$

$B = ?$

Sol. $g = 9.8$

m/L

$F = i(L \times B)$

$mg = iLB$

$B = \frac{m \times g}{i \times \cancel{L}} \Rightarrow \frac{0.0466 \times 9.8}{28\ \cancel{A}}$

$\boxed{B = 0.01631\ T}$

Problems Page 829

Q1)

Given Data

$\theta = 23^\circ$

$F_B = 6.50 \times 10^{-17}$

$B = 2.60 \, mT$

$V = ?$

$K.E = ?$

Sol

$F = qvB\sin\theta$

$$v = \frac{F}{qB\sin\theta} \Rightarrow \frac{6.50 \times 10^{-17}}{1.6\times10^{-19} \times 2.60\times10^{-3} \times \sin 23^\circ}$$

$$V = \frac{6.50}{1.625} \times 10^{-17+19+3}$$

$V = 4 \times 10^{5} \, ms^{-1}$

$\boxed{V = 400 \, kms^{-1}}$

mass of Proton

1.67×10^{-27}

b)

$$K.E = \frac{1}{2} mV^2$$

$$= \frac{1}{2} 1.67\times10^{-27} (400\times10^{3})^2$$

$K.E = 1.336 \times 10^{-16} \, J$

$1eV = 1.602\times10^{-19} J$

$$1\ J = \frac{1}{1.602 \times 10^{-19}}$$

$$= 6.242 \times 10^{18}\ eV$$

$$K.E = 1.33336 \times 10^{-16} \times 6.242 \times 10^{18}$$

$$\boxed{K.E = 835\ eV}$$

Q4)

Given Data

Not included

$V = 550\ ms^{-1}$

$B = 0.045T$

$q = 3.2 \times 10^{-19}$

$m = 6.6 \times 10^{-27}\ kg$

$\theta = 52^\circ$

$F_B = ?$

$a_c = ?$

speed = ?

Sol.

$$F_B = qVB \sin\theta$$

$$= 3.2 \times 10^{-19} \times 550 \times 0.045 \times \sin 52^\circ$$

$$= 62.410 \times 10^{-19}$$

$$F_B = 6.24 \times 10^{-18}\ N$$

b) $a = ?$

$F = ma$

$a = \frac{F}{m} \Rightarrow \frac{6.24}{6.6} \times \frac{10^{-18}}{10^{-27}}$

$= 0.945 \times 10^{-18+27}$

$= 0.945 \times 10^{9}$

$\boxed{a = 9.45 \times 10^{8}\ ms^{-2}}$

c)

speed remains same

Direction of B is $\perp$ to v.

Q.6)

Given Data

$\vec{B} = B_x(10\hat{i} - 20\hat{j} + 30\hat{k})\,mT$

$\vec{V} = V_x\hat{i} + V_y\hat{j} + (2\,km s^{-1})\hat{k}$

$\vec{F_B} = (4 \times 10^{-17} N)\hat{i} + (2 \times 10^{-17} N)\hat{j}$

$V_x = ?$

$V_y = ?$

$$\vec{B} \times \vec{V} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 10 & -20 & 30 \\ V_x & V_y & 2k \end{vmatrix}$$

$$= \hat{i}(-40k - 30V_y) - \hat{j}(40k - 30V_x)$$

$$+\hat{k}(10V_y + 20V_x)$$

$$= -40k\hat{i} - 30V_y\hat{i} - 40k\hat{j} + 30V_x$$

$$+ 10V_y\hat{k} + 20V_x\hat{k}$$

$$=$$

Q.7)

Given Data

$\vec{V} = 12\hat{j} + 15\hat{k}$

$a = 2 \times 10^{12}$

$\vec{B} = 400\mu T$

electron mass $= 9.11 \times 10^{-31}$

$E = ?$

Sol

~~$V = \frac{E}{B}$~~

~~$E = VB$~~

~~$= (12\hat{j} + 15\hat{k})\, 400\mu T$~~

~~$E = 400 \times 10^{-6} + 12 + 15$~~

$F = qe + qVB$

$=$

$F = ma$ — (1)

~~$F = qV$~~

$F = qe + qVB$ — (2)

compare

$ma = qE + qVB$

$ma = q(E + VB)$

$\frac{ma}{q} = E + VB$

$\frac{ma}{q} - VB = E$

$E = \frac{9.11 \times 10^{-31} \times 2 \times 10^{12}}{1.6 \times 10^{-19}} - (12\hat{j} + 15\hat{k})\, 400\mu T$

$j\times k$ $\cancel{k\times j}=i$ $\hat{k}$ j

$$E = \frac{18.22\times10^{-19}}{1.6\times10^{-19}} - (-4.8\times10^{-3} + 6\times10^{-3})$$

$$E = 11.38 + (4.8\times10^{-3}\hat{k} - 6\times10^{-3}\hat{j})$$

$$\boxed{E = 11.38\,i - 6\times10^{-3}\,\hat{j} + 4.8\times10^{-3}\,k} \ V/m$$

Q10)

Given Data

$B = -2.50\,\hat{i}\ mT$ $\qquad i\times i =$

$\vec{v} = 2000\,\hat{j}\ ms^{-1}$

$F_{net} = ?$ $\qquad i\times j = k$

$j\times k = i$

Sol.

a) $4.00\,\hat{k}\ V/m$ $\qquad k\times\hat{i} = j$

$j\times i = -k$

$F = qvB$ $\hat{j}$ $\hat{i}$

$= 1.6\times10^{-19}\times 2000\times -2.50\times10^{-3}$

$= +8000\times10^{-19-3}$

$$\boxed{F_B = +8\times10^{-19}\,\hat{k}\,N}$$

$F_E = qE$

$\Rightarrow 1.6\times10^{-19}\times 400\,\hat{k}$

$F_r = 6.4\times10^{-19}$

$$F_{net} = F_e + F_B$$

$$= \cancel{4.6}\,6.4 \times 10^{-19}\hat{k} + 8 \times 10^{-19}\hat{k}$$

$$= 14.4 \times 10^{-19}$$

$$\boxed{F_{net} \Rightarrow 1.44 \times 10^{-18}\ N}$$

b) b) = $-4.00\ V/m$

$$F = qvB$$

$$= 1.6 \times 10^{-19}$$

$$F = 8 \times 10^{-19}\ \hat{k}\ N$$

$$F = qE$$

$$= 1.6 \times 10^{-19} \times -4$$

$$F \Rightarrow -6.4 \times 10^{-19}$$

$$F_{net} = F_e + F_B$$

$$= -6.4 + 10^{-19} + 8 \times 10^{-19}$$

$$\cancel{F_{net}\ 1.44 \times 10^{-18}\ N}$$

$$\boxed{F_{net} = 1.6 \times 10^{-19}\ \hat{k}\ N}$$ Ans

c) $4.00\,\hat{i}\ V/m$

$$\boxed{F_B = 8\times10^{-19}\,\hat{k}\,N}$$

$F = qE$

$= 1.6\times10^{-19}\times4\,\hat{i}$ 4

$$\boxed{F_e \Rightarrow 6.4\times10^{-19}\,\hat{i}\ N}$$

Net force $F_e + F_B$

$F_{net} = \cancel{F_e + F} \Rightarrow qe + qvB$

$$\boxed{F_{net} = \left(6.4\times10^{-19}\,\hat{i} + 8\times10^{-19}\,\hat{k}\right) N}$$ Ans

$= \cancel{1.44\times10^{-19}\,\hat{j}}$

$l \Rightarrow$
$w \Rightarrow$
thick
$B =$
V
$v =$

$E = vB$ $\quad v = \frac{V}{Bd}$

$V = Ed.$ $\quad V = vBd.$

Q. 14)

Given Data

$l = 6.50\,cm = 0.065$

$w = 0.850\,cm = 8.5\times10^{-3}$

thick $= 0.760\,mm$

$B = 1.20\,mT$

$V = 3.90\,\mu V$

$v \Rightarrow$

$v = \frac{V}{Bd}$

$v = \frac{3.90\times10^{-6}}{1.20\times10^{-3}\times8.5\times10^{-3}}$

$\Rightarrow \frac{3.90}{10.2}$

$v = 0.382\,m/s$

17

Given Data

$q = 2e$

$m = 4u$

$1u = 1.661 \times 10^{-27} kg$

$r = 4.50 cm$

$B = 1.20 T$

v is $= ?$

$T = ?$

$K.E = ?$

potential diffence $= ?$

Sol

$r = \frac{mv}{qB}$

$v = \frac{r\, q\, B}{m}$

$\Rightarrow \frac{0.045 \times 3.2 \times 10^{-19} \times 1.20}{6.64 \times 10^{-27}}$

$= \frac{0.1728}{6.64} \times 10^{-19+27}$

$= 0.026 \times 10^{8}$

$= 2602409.6$

$v = 2.6 \times 10^{6}$ m/s

b) $T = \frac{2\pi m}{qB} \Rightarrow \frac{2\pi\, 6.64 \times 10^{-27}}{3.2 \times 10^{-19} \times 1.20}$

$= \frac{41.69}{3.84} \times 10^{-27+19}$

$= 10.85 \times 10^{-8} = 1.085 \times 10^{-7}$

$K.E = \frac{1}{2} mV^2$

$= \frac{1}{2}\, 6.64\times10^{-27}\, (2.6\times10^{6})^2$

$\boxed{K.E = 2.25\times10^{-14}}$

d)

$K = qV$

$V = \frac{K}{q} \Rightarrow \frac{2.25\times10^{-14}}{2\times1.6\times10^{-19}}$

$= 0.703\times10^{5}$

$= 70300$

$\boxed{V = 7\times10^{4}\ V}$

Q4)

Given Data

$L = 1.80\,m$

$B = 1.50\,T$

$i = 13\,A$

$\theta = 35^\circ$

Sol.

$F = iLB\sin\theta$

$F = 13\times1.80\times1.50\times\sin35^\circ$

$\boxed{F = 20.1\,N}$

Ch # 29

29-1) Megnetic field Due to a Current

$dB = \frac{\mu_0}{4\pi} \frac{i\,ds \sin\theta}{r^2}$ $\mu_0 = 4\pi \times 10^{-7}\ T\cdot m/A$
$\approx 1.26 \times 10^{-6}\ T\,m/A$

$$\boxed{d\vec{B} = \frac{\mu_0}{4\pi} \frac{i\,d\vec{s} \times \hat{r}}{r^2}}$$ ← Biot - savart law

BI

$$\boxed{B = \frac{\mu_0 i}{2\pi R}}$$

29-2 Force b/w 2 Parallel Plates

$F_{ba} = i_b L B_a \sin 90^\circ \Rightarrow \frac{\mu_0 L i_a i_b}{2\pi d}$

d is separation i_a and i_b are currents in the wires

29-3 Ampere's law

$$\boxed{\oint \vec{B} \cdot d\vec{s} = \mu_0 i_{enc}}$$

Megnetic Field of a long straight wire with current

$$\boxed{B = \frac{\mu_0 i}{2\pi r}}$$ outside wire

$$\boxed{B = \left(\frac{\mu_0 i}{2\pi R^2}\right) r}$$

Megnetic field inside the current carrying straigh wire?

$$\oint \vec{B}\cdot d\vec{s} = B \oint ds = B\, 2\pi r$$

$$i_{enclose} = i \frac{\pi r^2}{\pi r^2}$$

$$B(2\pi r) = \mu_0 i \frac{\pi r^2}{\pi R^2}$$

$$\boxed{B = \left(\frac{\mu_0 i}{2\pi R^2}\right) r}$$

29.4) Megnetic field of a Solenoid

$$\oint \vec{B}\cdot\vec{ds} = \int_a^b \vec{B}\cdot\vec{ds} + \int_b^c \vec{B}\cdot\vec{ds} + \int_c^d \vec{B}\cdot\vec{ds} + \int_d^a \vec{B}\cdot\vec{ds}$$

$$\oint \vec{B}\cdot\vec{ds} = Bh$$

$$i_{enc} = i(nh)$$

$$Bh = \mu_0 i$$

$$Bh = \mu_0 i(nh)$$

$\boxed{B = \mu_0 i n}$ infinitely Long long ideal Solenoid

29-4) Magnetic Field of a Toroid

$$\int B\, ds = \mu_0 i N$$

$$B\, 2\pi r = \mu_0 i N$$

$$\boxed{B = \frac{\mu_0 i N}{2\pi r} \quad \frac{1}{r}}$$

Example 29.03

Given Data

$a = 2\,cm$ $\qquad$ 3.9×10^{-7}

$b = 4\,cm$

$J = cr^2$

$c = 3.0\times10^{6}\,A/m^4$ $\qquad$ $B = \mu i$

$r = 3\,cm$

$\vec{B} = ?$ $\qquad$ cylinder area $= 2\pi r l$

Sol:-

$$i = \int_a^r J\,dA \Rightarrow J\int_a^r 2\pi r\,dr \Rightarrow \cancel{cr^2} cr^2 \int_a^r 2\pi r\,dr$$

$$= 2\pi c\int_a^r r^3\,dr \Rightarrow 2c\pi\left[\frac{r^4}{4}\right]_a^r$$

$$= \frac{2c\pi}{4}\left[r^4 - a^4\right] \Rightarrow \frac{1}{2}\pi c\,(r^4 - a^4)$$

$$i = \frac{1}{2}\pi c\,(r^4 - a^4)$$

$$B = \mu i \Rightarrow \mu\,\frac{1}{2}\,\pi c\,(r^4 - a^4)$$

$$= 4\pi\times10^{-7}\times\frac{1}{2}\,\pi\,3.0\times10^{6}\left((0.03)^4 - (0.02)^4\right)$$

$$= 0.6 \qquad (8.1\times10^{-7} - 1.6\times10^{-7})$$

$$\oint \vec{B} \cdot \vec{ds} = \mu_0 i$$

$$B\, 2\pi r = -\frac{\mu_0 \pi C}{2} (r^4 - a^4)$$

$$B = -\frac{\mu_0 C}{4r} (r^4 - a^4)$$

$$= \frac{4\pi \times 10^{-7} \times 3.0 \times 10^{+6} ((0.03)^4 - (0.02)^4)}{4 (0.03)}$$

$$= \frac{0.942}{0.03} (8.1 \times 10^{-7} - 1.6 \times 10^{-7})$$

~~31.4 (~~

$$\boxed{B = 2.041 \times 10^{-5}}$$

Example 29.04

Given Data

$L = 1.23 m$

$i = 5.57 A$

$d = 3.55 cm$

$N = 850$ tuns

Total turns $= 5$

Sol $B = ?$

$$B = \mu_0 i n \Rightarrow 4\pi \times 10^{-7} \times 5.57 \times \frac{5 \times 850}{1.23}$$

$= 241728 \Rightarrow$ ~~2.4 × 10~~

$= 2.24 \times 10^{-2} \Rightarrow$ ~~2.4~~ $24.2 \times 10^{-3} T$

P ch # 29

Pooblems Pg # 856

1)

Given Data

$R = 6.1m$

$i = 100A$

$B = 20\mu T$

$B = \frac{\mu i}{2\pi R}$

Sol

$B = \frac{4\pi \times 10^{-7} \times 100}{2\pi \quad 6.1} \Rightarrow \frac{\cancel{1.256}\ 2 \times 10^{-5}}{6.1}$

$B = 0.327 \times 10^{-5}$

$\boxed{B = 3.3 \times 10^{-6}\ T}$

b) $\frac{3.3\ \mu T}{20\ \mu T} \Rightarrow 16.5$ percent

Therefore, this could interfere with the compass reading.

Q.351

$i_A = 4mA$

$d_1 = 2.40cm = 0.024$ $\qquad B = \frac{\mu_0 i_A i_B l}{2\pi d}$

$d_2 = 5cm = 0.05$

$i_B = 6.80mA$

$\frac{F}{L} = ?$

$\theta = \tan^{-1} \frac{2.40\,cm}{5\,cm}$ $\qquad \boxed{\theta = 25^\circ}$

$d = \sqrt{5.76\times10^{-4} + 2.5\times10^{-3}}$

$d = 0.055m$

$F = \frac{\mu_0 l L i_A i_B}{2\pi d} \cos\theta$

$= \frac{4\pi\times10^{-7}\ 0.024\times0.05}{2\pi\ (0.055)} \cos 25^\circ$

$\frac{F}{L} = \frac{2.4\times10^{-10}}{0.055} \cos 25^\circ$

$\boxed{\frac{F}{L} = 8.84\times10^{-11}\ N/m}$

$$\frac{F}{L} = -\frac{\mu_0 i_A i_B}{2\pi d} \qquad \frac{2\pi d}{\mu_0 i_A} = i_B$$

Q45)

Given Data

1

$i = 2A$

Sol

$$\int B \cdot ds = \mu i$$

$$= 4\pi \times 10^{-7} (2-2-2)$$

$$\boxed{\int B \cdot ds = -2.5 \times 10^{-6}\ Tm}$$

b)

$$\int B \cdot ds = \mu_0 i$$

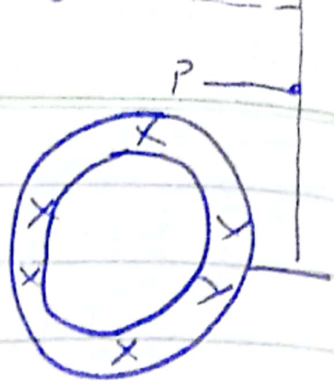
$$= 4\pi \times 10^{-7} (2-2+2-2)$$

$$\boxed{\int B \cdot ds = 0}$$

Imp

48

Given Data



$R = 2.6\,cm$

$i = 8\,mA$

$R = 3R$

$\frac{i_0}{6_3} = \frac{l}{r}\left(\frac{i_0}{2} - 1\right)$

Sol.

$B = ?$

$B_c = B_p + B_{wire}$

$B_c = 0 + \frac{\mu_0 i}{2\pi(3R)} \Rightarrow \frac{\mu_0 i}{6\pi R}$

$B_p = B_p + B_{wire}$

$= \frac{\mu_0 i_0}{2\pi(2R)}\hat{j} + \frac{\mu_0 i}{2\pi(2R)}(-\hat{j})$

$B_p = \frac{\mu_0}{2\pi R}\left(\frac{i_0}{2} - i\right)-\hat{j}$

$\frac{i_0 + 3i}{3}$

Compare B_p and B_c

$\frac{i}{3} = \frac{i_0}{3} - \frac{3i}{3}$

$\frac{i}{3} + i = \frac{i_0}{3}$

$\frac{\cancel{\mu_0} i}{\underset{3}{\cancel{6}}\cancel{\pi R}} = \frac{\cancel{\mu_0}}{\cancel{2\pi R}}\left(\frac{i_0 - 2i}{2}\right)$

$\frac{i + 3i}{3} = \frac{i_0}{3}$

$i + 3i = i$

~~$\frac{\mu_0 i}{3} = \frac{i_0 - 3i}{3}$~~

$\frac{i}{3} = \frac{i_0 - 2i}{2}$

$\frac{i}{3} = \frac{i_0}{2} - i \rightarrow \frac{i}{3} + i = \frac{i_0}{2}$

$\frac{i + 3i}{3} = \frac{i_0}{2}$

$$\frac{\mu_0 i_1}{2\pi d\cos 45} + \frac{\mu_0 i_2}{2\pi d\cos 45}$$

$$2\,4i = \frac{3i_0}{2}$$

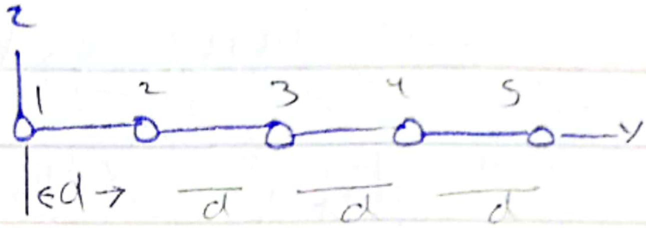
$$i = \frac{3i_0}{8} \Rightarrow \frac{3i}{8} \Rightarrow \frac{3\times 8\times 10^{-3}}{8}$$

$$i = \frac{3\times \cancel{8}\times 10^{-3}}{\cancel{8}}$$

$$\boxed{i = 3mA}$$

b) Direction is into the page.

Q39) Given Data



$i_1 = 2A$

$i_3 = 0.250A$, $d = 50cm \Rightarrow 0.50m$

$i_4 = 4A$, $d = 0.50m$

$i_5 = 2A$

$i_2 = 4A$ ×

$\frac{F}{l} = ?$

Sol $F = \frac{\mu_0 i_1 i_2 l}{2\pi d}$

$$\frac{F}{l} = \frac{\mu_0 i_3}{2\pi d}\left[-\frac{i_1}{2d} + \frac{i_2}{d} + \frac{i_4}{d} + \frac{i_5}{2d}\right]$$

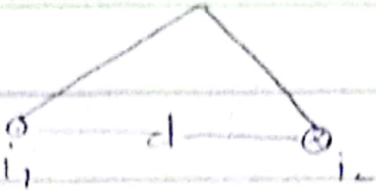
$$= \frac{4\pi\times 10^{-7}\times 0.250}{2\pi}\left[-\cancel{2} + 0.50 + 8 + \cancel{2}\right]$$

$$\Rightarrow \cancel{5\times 10^{-8}} + \frac{9\times 10^{-8}}{0.50} \Rightarrow \boxed{8\times 10^{-8}}$$

$$\boxed{\frac{F}{l} = 8\times 10^{-8}\,N/m}$$

Q 29.2

Given Data



$i_1 = 15A$

$|B|$

$i_2 = 32A$

$d = 5.3cm \Rightarrow 0.053$

Sol

$B_1 = \frac{\mu_0 i}{2\pi d} \cos 45^\circ \quad B_2 = \frac{\mu_0 i}{2\pi d} \cos 45^\circ$

$B = \sqrt{B_1^2 + B_2^2}$

$= \sqrt{\left(\frac{\mu_0 i_1}{2\pi d}\right)^2 + \left(\frac{\mu_0 i_2}{2\pi d}\right)^2} \Rightarrow \sqrt{\frac{\mu_0^2 \theta_1}{4\pi^2 d^2}\left(i_1^2 + i_2^2\right)}$

$= \frac{\mu_0}{\cos 45^\circ 2\pi d}\sqrt{i_1^2 + i_2^2} \Rightarrow \frac{4\pi \times 10^{-7}}{2\pi (0.053)}\sqrt{(15)^2 + (32)^2}$

$= \frac{2\times 10^{-7}}{0.053}\frac{(38)}{(0.707)} \Rightarrow \frac{7.6\times 10^{-6}}{0.037471} \Rightarrow 2\times 10^{-4}\,T$

$\phi = \tan^{-1}\frac{B_1}{B_2} \Rightarrow \boxed{\phi = 25^\circ}$

$\cancel{45^\circ - \phi = 25^\circ}$ 0 $\phi = 45 + 25 = 70^\circ$